

Arindam Shiva Tripathi

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EDUCATION

University of Maryland (UMD)

College Park, MD, USA

Master of Science in Human-Computer Interaction (HCIM) — GPA 4.0

Aug 2025 – May 2027

- M.S. Thesis (in progress), building on Reframed; advised by Dr. Ming Lin and Dr. Beth Bonsignore

National University of Singapore (NUS)

Singapore

Bachelor of Engineering in Computer Engineering, Minor in Business Management

Aug 2019 – Jul 2023

- Concentration in Interactive Digital Media; graduated with honors

EXPERIENCE

LIGHT Helmets

Carlsbad, CA (Remote)

3D Scanning & Fitment Software Intern

May – Aug 2026

- Sole developer of an end-to-end scan-to-fitment pipeline for two helmet families, turning 3D head and helmet-interior scans into per-station padding recommendations for ideal fitment
- Built an Android head-scanning app using ARCore to capture 3D head geometry for the fitment pipeline
- Ran a 4-axis, derivative-free grid search for automated helmet seating, outperforming a coordinate-descent baseline

L'Oréal

Singapore

Data Analytics Specialist (Full-Time)

Jul 2023 – Aug 2025

- Led 2 developers to build a suite of ML-backed planning tools from proof-of-concept to production, used by 200+ analysts across 10+ APAC markets and improving promotional efficiency by 24%
- Designed and iterated on analytics tools with stakeholders from the ground up, lifting analytics adoption by 40%

SP Digital

Singapore

Full Stack Developer Intern

Jun 2021 – Dec 2021

- Built responsive React/JavaScript features for the SP Digital app (1.2M+ downloads)

PROJECTS

Reframed — Quest 3 Passthrough AR (M.S. Thesis)

Mar 2026 – Present

Unity, C#, OpenCV

- Independently developed during UMD's New Works Incubator, Reframed is an interactive AR experience where classic artwork animates and extends beyond its frame using AR and becomes a larger treasure hunt through the art
- Built a Quest 3 passthrough AR system that recognizes physical artworks in real space and anchors animated digital content to each one, using OpenCV, ORB feature matching, homography, and solvePnP pose estimation
- Showcased the project at UMD's NextNOW Fest, drawing 50+ visitors over two days to strong, positive reactions

Niantic Spatial AR — XR Club Project Lead & Incoming Board Member

Mar 2026 – Present

Unity, C#, Niantic Spatial SDK

- Selected for a competitive university-Niantic partnership, leading a 3-person team building location-aware mobile AR

Aura Desk — Ambient AR Productivity Surface

Jun 2026 – Present

Android XR, Jetpack XR SDK, Kotlin, On-Device AI

- Built an MVP on unreleased Xreal Project Aura smart glasses that turns a physical desk into an ambient AR computing surface, with glanceable widgets for tasks, calendar, and music anchored in the user's space
- Integrated on-device AI and voice commands so the surface handles natural-language requests hands-free
- Became a Google Trusted Developer for early platform access; prototyping wireless laptop screen extension next

SnaPT — ChatGPT Context-Management Extension

Apr 2026

JavaScript, Chrome Extension APIs, Material 3

- Shipped a live Chrome extension that keeps long ChatGPT threads responsive, reverse-engineering the app's conversation payload to trim active context before render while keeping older turns recoverable

SKILLS

Languages: C#, Python, Kotlin, JavaScript, TypeScript, C/C++

XR & Spatial: Unity, Android XR, Jetpack XR SDK, Meta XR SDK, OpenXR, AR Foundation, ARCore, Niantic SDK

3D, Graphics & Vision: Three.js, WebGL, Blender, Open3D, trimesh, OpenCV (ORB, homography, pose estimation)

Prototyping & Hardware: TouchDesigner, React, Figma, Raspberry Pi, ESP32

Certifications: Google Trusted Developer, Android XR (2026); Google Cloud Certified Cloud Digital Leader (2023)